

Project Description: Regenerative AI for Sustainable Agriculture

The **Smart Crop Monitoring Framework** is a comprehensive, AI-driven solution designed to advance precision agriculture and empower farmers with accessible technology. This project stands at the intersection of machine learning, remote sensing, and web development to provide actionable insights into crop health. It aims to tackle critical agricultural challenges by offering a dual-pronged approach to monitoring: high-resolution plant disease detection and broad-scale field health assessment.

Core Functionality

The framework's power lies in its two primary, integrated modules:

1. **AI-Powered Plant Disease Detection:** At the micro-level, the project employs a lightweight **Convolutional Neural Network (CNN)** to identify plant diseases from leaf images. Trained on the extensive **PlantVillage dataset**, the model can accurately classify various conditions, enabling farmers to perform early diagnosis. This feature allows for timely intervention, which can prevent the spread of disease, reduce crop loss, and minimize the need for broad-spectrum chemical treatments.
2. **Satellite-Based Field Health Analysis (NDVI):** At the macro-level, the framework leverages remote sensing data from **Sentinel-2 satellites**. It processes the Red (B04) and Near-Infrared (B08) spectral bands to calculate the **Normalized Difference Vegetation Index (NDVI)** using the formula: $NDVI = \frac{(NIR - Red)}{(NIR + Red)}$. The resulting NDVI values are visualized as an intuitive **heatmap**, providing a clear overview of the entire field's health. This allows for the identification of areas under stress due to issues like water scarcity, nutrient deficiency, or widespread disease, enabling targeted resource management.

Technology and Implementation

This project is a showcase of a modern data science stack applied to a real-world problem:

- **Machine Learning:** **TensorFlow** is used to build and train the CNN model for disease detection. The repository includes the pre-trained model (`plant_disease_cnn_model.h5`) for immediate use and a Jupyter Notebook (`train_plant_disease_cnn.ipynb`) for full transparency and reproducibility.
- **Web Application:** A user-friendly interface is built with **Streamlit**, allowing users to simply upload images (crop leaves or satellite GeoTIFF files) and receive instant predictions and visualizations. This makes the sophisticated backend models accessible to a non-technical audience.
- **Data Processing:** The framework uses **Rasterio** to handle geospatial satellite data for NDVI calculations and libraries like **NumPy** and **Pillow** for numerical and image manipulation.
- **Deployment:** The architecture is designed with both cloud and edge computing in mind. While it runs as a web app, its components are optimized for potential deployment on edge devices like the Jetson Nano, enabling offline inference in remote areas with limited internet connectivity.

Project Significance and Future Vision

Rooted in academic research, this project is the practical implementation of the paper "Regenerative AI for Sustainable Agriculture: A Smart Crop Monitoring Framework." The

repository is meticulously structured, with clear installation instructions, dependency management (requirements.txt), and scripts for data preparation, making it an excellent resource for other researchers and developers.

The project also has a clear and ambitious roadmap for future development, including:

- Integration with **IoT soil sensors** for more granular data.
- Adding a **weed detection module** using advanced models like YOLOv8.
- Enhancing accessibility with **multi-language support** for a global farming community.

In summary, the Smart Crop Monitoring Framework is a powerful and well-documented project that not only provides a valuable tool for sustainable agriculture but also serves as an exemplary end-to-end data science application.

GITHUB REPO LINK: <https://github.com/kirtivardhan80/smart-crop-monitoring.git>